

The direction of self-evaluation bias in active learning depends  
on the metric:  
accuracy underestimates, macro-F1 overestimates

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### Abstract

Practitioners of pool-based active learning routinely evaluate their model on a held-out split of the labels they have collected — because that is all they have. It is folklore that this self-evaluation is biased, since actively sampled data are not i.i.d.; but the direction and size of the bias are rarely measured. We measure both, at scale, with a perfect-oracle protocol that separates a labelable *pool* (50,000 short product descriptions, 649 classes, severe long tail) from a *reserved population* (181,490 instances) never touched by selection. At each step the collected labels are split 80/20; the same model is scored on the internal test split and on the population. Across five selection policies and two classifiers we find the bias is large, systematic and *metric-dependent*: under uncertainty sampling, internal accuracy *underestimates* deployment accuracy by  $17 \pm 1$  points across eight seeds (the internal test inherits the hard cases), while internal macro-F1 *overestimates* it by up to 34 points (the active sample over-represents rare classes). A random-sampling control shows near-zero gap, validating the instrument; entropy beats random in ceiling on 8 of 8 seeds (paired Wilcoxon,  $p = 0.0078$ ). Two corollaries follow. First, *more labels can hurt*: for a logistic-loss classifier, training on the  $\approx 15k$  actively selected labels beats training on the full 50k pool by 15 macro-F1 points, because the active sample acts as a balanced curriculum — while a per-class-normalized prototype classifier is immune. Second, an ablation of a semantic-diversity selection policy shows its entire continuous-regime gain comes from stratifying by predicted class, yielding a trivial, strong, model-agnostic baseline. Self-evaluated metrics from active learning runs should not be used for deployment decisions in either direction; we quantify why, and what to do instead.

**Keywords:** active learning; sampling bias; evaluation; class imbalance; macro-F1; text classification.

## 1 Introduction

Pool-based active learning (AL) buys labels selectively (Settles, 2012; Lewis and Gale, 1994). Its success metric — performance per label — is usually reported on a benchmark with abundant held-out gold data. Deployed AL has no such luxury: the only labeled data available for evaluation are the labels the loop itself collected, and the standard practice is to hold out a fraction of them.

The resulting sample is not i.i.d. by construction: uncertainty sampling over-represents boundary cases and rare classes (Attenberg and Provost, 2010; Santos, 2016). That this biases evaluation is known in principle; *how much, in which direction, and how the direction interacts with the metric* is rarely quantified — yet these are exactly the questions a practitioner needs answered before trusting an internally computed number for a release decision.

We answer them with a deliberately simple protocol (Section 3): a perfect oracle (gold labels), a fixed labelable pool, and a large reserved population that selection can never touch. The gap between internal and population scores, tracked along the full labeling trajectory, *is* the self-evaluation bias — and a random-sampling control arm certifies that the gap is caused by selection, not by the instrument.

Contributions: (i) the first (to our knowledge) large-scale, controlled quantification of AL self-evaluation bias with 649 classes, showing it is **bidirectional and metric-dependent** — internal accuracy underestimates deployment performance while internal macro-F1 overestimates it; (ii) a mechanistic account and empirical demonstration of the *less-is-more* effect — full-pool labeling can be worse than stopping at 30% for loss functions dominated by the natural class distribution — including a classifier (per-class prototype) immune by construction; (iii) an ablation showing a semantic-diversity continuous policy owes its gain entirely to prediction-stratification, exposing a trivial strong baseline; (iv) operational guidance: what internal metrics can and cannot support, and a reserved-set release criterion.

## 2 Related work

Sampling bias in AL is acknowledged since the field’s foundations (Settles, 2012); failure modes of uncertainty sampling under imbalance are documented (Attenberg and Provost, 2010), and bias-aware AL has been studied theoretically (Santos, 2016). Density and cluster-based policies (Settles and Craven, 2008; Nguyen and Smeulders, 2004; Dasgupta and Hsu, 2008) motivate the diversity component we ablate. Metric behavior under imbalance (micro vs. macro aggregation) is classical (Sokolova and Lapalme, 2009). What is missing in this literature — and what we contribute — is a controlled, trajectory-level measurement of the *evaluation* bias (not the training bias) with its metric dependence, at a class cardinality where the two metrics dissociate dramatically. The renewed practical interest comes from LLM-oracle pipelines (Gilardi et al., 2023; Bayer and Reuter, 2024), where cheap labels make full-trajectory experiments affordable and self-evaluation is the default telemetry.

## 3 Protocol: internal vs. population evaluation

Let  $D$  be a deduplicated corpus, shuffled once with a fixed seed and split into a **pool**  $P$  ( $|P| = 50,000$ ), from which the learner may request labels, and a **population**  $Q$  ( $|Q| = 181,490$ ), reserved in full. The oracle is perfect (gold labels), removing annotation noise as a confounder. The loop starts from a random batch and iterates: train on the collected set  $L_t$ , select the next batch of 500 from  $P \setminus L_t$ , until the pool is exhausted. At every step,  $L_t$  is split 80/20 (stratified when feasible, fixed seed); the model is trained on the 80% and scored twice with identical weights: on the **internal test** (the 20% — what a practitioner without reserved data can measure) and on the **population**  $Q$  (deployment performance). The signed gap  $\Delta_m(t) = m_{\text{int}}(t) - m_{\text{ext}}(t)$  for metric  $m$  is the self-evaluation bias. Both scores use the same computation convention: macro-F1 averages over the classes present in the respective reference set, which is precisely the

support asymmetry a practitioner faces — the internal test can only contain collected classes. Because the internal test is small early in the trajectory (20% of  $|L_t|$ ), single-step gaps are noisy; all reported biases are trajectory-level patterns sustained over many consecutive steps, not single-point readings.

**Control arm.** Under random selection,  $L_t$  is an i.i.d. sample of  $P$  (and  $P$  of  $D$ ), so  $\Delta$  should vanish up to noise. A non-zero  $\Delta$  under random selection would indicate an instrument artifact (split leakage, size effects); observing  $\Delta \approx 0$  validates the protocol, and every deviation under a non-random policy is then attributable to selection.

## 4 Experimental setup

**Data and task.** 231,490 deduplicated short product descriptions from Brazilian fiscal receipts (4–50 characters, uppercase, abbreviations), 649 classes with at least 2 instances, severe long tail. **Classifiers**, chosen as extremes of loss geometry: *SGD* — logistic regression on binary TF-IDF via stochastic gradient descent, loss dominated by the empirical class distribution; *PVBin* — a per-class prototype model over binary term vectors (Darú, 2024), normalized per class by construction. **Policies**: uncertainty (entropy); random (control); a semantic-diversity policy (density-representative selection over SBERT embeddings with lexical-novelty tie-breaking, DRI-SL); the same policy with groups given by the classifier’s *predicted classes* (DRI-SL-C); and the ablation of the latter without lexical novelty — equivalent to *stratified sampling by predicted class*. Batch 500, budget to pool exhaustion. The entropy and random arms (the paper’s central comparison) were run over **eight seeds** that vary the initial draw, tie-breaks and internal split while holding the pool fixed; results for these arms are reported as mean $\pm$ SD with a paired Wilcoxon test across seeds. The diversity variants (DRI-SL, DRI-SL-C and its ablation) were run single-seed and are reported descriptively.

## 5 Results

### 5.1 Selector comparison on the population

Table 1: Population macro-F1 by policy and classifier. Entropy and random over 8 seeds (mean $\pm$ SD); diversity variants single-seed (descriptive). Saturation: smallest  $|L|$  reaching 95% of the arm’s own ceiling.

| Clf.  | Policy                 | Ceiling                           | Saturation                      | F1@10k | F1@20k |
|-------|------------------------|-----------------------------------|---------------------------------|--------|--------|
| SGD   | Entropy                | <b>0.592<math>\pm</math>0.004</b> | <b>9,060<math>\pm</math>560</b> | 0.565  | 0.574  |
| SGD   | Stratif. by prediction | 0.555                             | 10,000                          | 0.533  | 0.543  |
| SGD   | DRI-SL-C               | 0.491                             | 15,500                          | 0.412  | 0.486  |
| SGD   | Random                 | 0.455 $\pm$ 0.004                 | 15,500 $\pm$ 1,100              | 0.391  | 0.449  |
| SGD   | DRI-SL                 | 0.441                             | 41,500                          | 0.310  | 0.365  |
| PVBin | Entropy                | 0.531 $\pm$ 0.002                 | 19,310 $\pm$ 700                | 0.438  | 0.509  |
| PVBin | Stratif. by prediction | 0.528                             | 18,000                          | 0.444  | 0.498  |
| PVBin | DRI-SL-C               | 0.525                             | 39,500                          | 0.349  | 0.453  |
| PVBin | Random                 | 0.528 $\pm$ 0.004                 | 40,630 $\pm$ 1,410              | 0.348  | 0.423  |
| PVBin | DRI-SL                 | 0.527                             | 45,500                          | 0.284  | 0.356  |

Entropy dominates at scale (Table 1), and the eight-seed campaign shows the dominance is not a lucky draw: for SGD it saturates at  $9,060 \pm 560$  labels (range 8,500–10,000) against  $15,500 \pm 1,100$  for random, and the entropy ceiling exceeds random’s in **8 of 8 seeds** (paired Wilcoxon, the maximum significance reachable with eight pairs,  $p = 0,0078$ ); for PVBIn the entropy edge in ceiling is small and not significant (6/8 seeds), but its saturation ( $19,310 \pm 700$ ) is still less than half of random’s ( $40,630 \pm 1,410$ ). The semantic-diversity policy (DRI-SL), designed for the zero-label cold start, is *worse than random* as a continuous policy; guiding its groups by predicted classes (DRI-SL-C) repairs it (ceiling  $0.441 \rightarrow 0.491$  for SGD, saturation  $2.7\times$  earlier).

## 5.2 The ablation: stratification is the active ingredient

Removing the lexical-novelty term from DRI-SL-C — leaving plain stratified sampling by predicted class — *improves* it on both classifiers (SGD: ceiling 0.555 vs. 0.491, saturation 10k vs. 15.5k; PVBIn: saturation 18k vs. 39.5k, tying entropy). The diversity heuristic, valuable at cold start where no model exists, is counterproductive in the continuous regime: it privileges lexically atypical exemplars when the macro metric pays for balanced class coverage. The practical corollary is a baseline: *predict, stratify, sample* — model-agnostic, no embeddings, no tuning — that captures most of the uncertainty-sampling gain and nearly all of the diversity-policy gain.

## 5.3 The bias is bidirectional and metric-dependent

The control behaves: under random selection the internal–external accuracy gap is  $\Delta \approx 0$  along the whole trajectory. Under entropy selection, internal **accuracy underestimates** the population value substantially and *robustly across seeds* — at  $|L| = 10,000$  the gap is  $-17,1 \pm 1,0$  points over the eight seeds (tight SD): the internal test inherits the boundary-heavy, hard-case distribution that uncertainty sampling selects. Internal **macro-F1 overestimates** the population value for *every* non-random policy — by up to +34 **points** early in the DRI-SL trajectory (single seed): the active sample gives rare classes internal test support far above their population share, so per-class scores are computed on flattering supports. Neither direction is usable as a deployment estimate, and the two errors do not cancel: a practitioner reading internal accuracy will under-deploy (the model is better than it looks); one reading internal macro-F1 will over-promise tail quality (it is worse than it looks).

## 5.4 More labels can hurt: the active sample as curriculum

For SGD under entropy selection, population macro-F1 peaks at 0.59 around 15k labels and *falls* to 0.44 when the entire pool is labeled: the final 35k labels, drawn increasingly from the natural (imbalanced) distribution, drag the logistic loss toward the head classes. The actively selected subset acts as a balanced curriculum — and labeling everything undoes it. PVBIn, whose per-class prototypes normalize away the class distribution, is immune (ceiling  $\approx$  flat across policies). Beyond waste, additional labels can be *harmful* to macro metrics for distribution-sensitive losses; remedies are early stopping by stagnation, reweighting, or per-class-normalized models.

# 6 Operational guidance

(i) Maintain a reserved evaluation set from day one — the bias cannot be corrected post hoc from internal data alone, since its size and sign depend on the (policy, metric, model) triple. (ii) Use

internal metrics only for *relative* decisions inside one run (stagnation-based stopping), never for absolute release decisions. (iii) If a reserved set is impossible, prefer internal accuracy over internal macro-F1 for go/no-go: it errs conservative. (iv) For macro objectives with distribution-sensitive losses, treat the labeled-set composition as part of the model: stopping earlier, subsampling or reweighting may beat labeling more.

## 7 Limitations

Single dataset and domain (short retail text; the mechanism — support inflation of rare classes — is general, but effect sizes are not); the central entropy-vs-random comparison is inferential over eight seeds, but the diversity-variant arms (DRI-SL, DRI-SL-C, ablation) remain single-seed and descriptive, so their between-arm orderings should be read as indicative pending a seed campaign of their own; perfect oracle by design (noisy oracles would add a second bias source; our companion work on LLM oracles addresses it separately, keeping this paper’s question clean).

## 8 Conclusion

Self-evaluation on actively collected data is not merely noisy — it is systematically wrong in a direction that depends on the metric: accuracy reads low, macro-F1 reads high, by double-digit margins at 649-class granularity. A random control certifies the effect is caused by selection. The same experiments yield a practical bonus (stratified-by-prediction as a trivial strong policy) and a warning (labeling the full pool can damage macro performance for distribution-sensitive losses). All trajectories, code and per-step artifacts are released.

**Artifacts.** [github.com/GHDaru/activelearning](https://github.com/GHDaru/activelearning) (experiments/e6population).

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